Sonargaon University

Course Title: ***Entrepreneurship Development***
Course Code: ***BA 2206***
Report On: Bottled water Business plan

Submitted To:
Md. Mostakin
Lecturer, BBA
Sonargaon University
147/1, Green Road, Panthapath, Dhaka

Prepared by The Innovators

Member Name	Student ID	Role
RASEL SHIKDER	BBA2402032084	Leader
ABU HURAYRA AL ABID	BBA1903018090	Information Collector
TAMIM HASAN SAMI	BBA2402032140	Researcher
MD. SOYEEB SHIKDER	BBA2402032180	Researcher
SAIDUL ISLAM	BBA2402032154	Information Collector

Date of Submission : 16th November, 2025

Savior Bottled Water: A Strategic Investment Report on Value Leadership and Sustainable Operations in Bangladesh

I. Executive Summary

This report provides a comprehensive analysis of the market opportunity, operational strategy, and financial projections for Savior Bottled Water, a new brand entering the Bangladesh mass market. Savior is strategically positioned to address two compounding crises: the widespread lack of access to affordable, reliable, purified drinking water and the escalating environmental catastrophe driven by single-use plastic waste in the country.

Savior's core thesis relies on leveraging advanced multi-stage purification technology, including Reverse Osmosis (RO) and Ultraviolet (UV) sterilization, to deliver certified purity. This premium quality water is made accessible through an aggressive penetration pricing strategy, offering products at prices significantly lower than established mass-market competitors such as Mum and Kinley. The initial product line will focus on 250ml, 500ml, and 1L bottles, with rapid expansion planned for 2L and 5L sizes to capture the essential household and business-to-business (B2B) markets.

Operationally, the brand is structured for sustainability and cost efficiency. This includes employing optimized system engineering, such as the application of Calcium Carbonate Precipitation Potential (CCPP) analysis to maximize RO permeate recovery and reduce water wastage, alongside the commitment to utilizing solar energy for production. The sustainable packaging strategy balances the use of plant-based materials with the promotion of high-durability, reusable plastic formats, directly addressing the endemic failure of local waste management infrastructure.

While the initial test phase financial projection indicates a net monthly operating loss of 10,500 BDT, resulting from the absorption of fixed costs over low initial volume , the financial model demonstrates a clear and achievable path to profitability. Break-even is expected after doubling the initial production scale, a target supported by the brand's highly competitive pricing (14–16 BDT for 500ml) and the significant emerging market demand for affordable, certified safe water. Savior is designed as a value brand that delivers measurable positive environmental and social impact, establishing a profitable business model aligned with global sustainability mandates.

II. The Dual Crisis: Water Scarcity and Plastic Pollution in Bangladesh

A. Introduction to the Problem: Accessing Clean Drinking Water

The demand for bottled water in Bangladesh is driven not by luxury preferences but by a profound public health imperative resulting from compromised public water sources. While a broad definition suggests water coverage of 87% in 2019, the supply remains intermittent and frequently unsafe. Groundwater, historically the primary resource for both drinking and irrigation, has become dangerously contaminated in large parts of the country, prompting consumers to seek reliable, safe alternatives.

The most critical systemic threat is arsenic contamination, a major public health disaster first identified in tubewell water in 1993. Tubewells are the main source of drinking water in rural areas, and it is estimated that 50 million people in Bangladesh are at risk of chronic exposure to arsenic through contaminated water consumption. Chronic exposure leads to severe consequences, collectively termed arsenicosis, including dermatological manifestations, multi-organ pathologies, and an increasing incidence of noncommunicable diseases and cancers. Furthermore, studies have documented elevated concentrations of other contaminants, such as Manganese (Mn) and Boron (B), with certain samples exceeding World Health Organization (WHO) limits. These elevated concentrations contribute to higher chronic risk values, indicating that children are particularly vulnerable to harm through oral exposure to drinking water.

Because the consumption of contaminated water must be halted to prevent and manage arsenicosis, the need for safe, purified water is essential for public health and well-being. Traditional mitigation efforts require solutions that are cheap, easy to use, and locally maintainable. Bottled water, when purified and certified, ceases to be a discretionary purchase and becomes a necessary health commodity. This essential need creates a massive, inelastic demand segment in Bangladesh, provided the product can be delivered affordably. Savior's challenge, therefore, is to convert this critical health requirement into an accessible, high-volume product, strategically differentiating itself from high-priced brands like Kinley, which benefit primarily from pre-existing consumer trust but remain inaccessible to the mass market.

B. Environmental Issue with Plastic Waste: Focusing on Bangladesh

The proliferation of bottled water, driven by the health crisis, simultaneously fuels a staggering environmental catastrophe in Bangladesh concerning single-use plastic (SUP) waste. Quantifying the problem reveals a dire disposal failure rate: Bangladesh generates between 3.15 and 3.84 billion SUP bottles annually. A monumental 78.6% of these bottles are not

recycled, contributing significantly to the pollution of rivers, oceans, and overburdened landfills.

The recycling infrastructure is fundamentally inadequate to cope with this volume. Only 21.4% of SUP bottles are recycled nationwide. In Dhaka, which generates over 30% of the nation's total waste, only 37.2% of plastic waste is recycled. The visible environmental consequences are severe. Major waterways, including the Buriganga, Turag, and Shitalakshya Rivers, are heavily polluted with plastic debris, disrupting river dredging activities, devastating aquatic ecosystems, and impacting the livelihoods of fishing and tourism industries. Furthermore, plastic pollution exacerbates climate and infrastructure risks, as clogged drainage systems increase the frequency and severity of urban flooding.

The long-term effects of this environmental mismanagement are concerning. Single-use plastic bottles can persist in the environment for up to 450 years, degrading into microplastics and releasing toxic chemicals such as Bisphenol A (BPA). These substances bio-accumulate in ecosystems, increasing toxicity in food chains and posing a chronic threat to human health through contamination.

A critical element influencing the disposal failure rate is the profound lack of public awareness regarding the risks. While 83.6% of Bangladeshis regularly use SUP bottles, risk awareness is critically low: only 18.4% of urban and 5.5% of rural respondents recognize the potential health and environmental hazards posed by the plastic containers. This prevalent disregard for proper disposal, coupled with inadequate infrastructure, results in consumers indiscriminately discarding bottles after a single use. This systematic lack of understanding underscores the necessity for Savior’s eco-strategy to go beyond mere use of "eco-friendly" materials. The strategy must heavily center on active consumer education, transparent communication, and promoting active reuse programs, recognizing that passive composting or recycling systems cannot be relied upon due to current infrastructural deficits and consumer habits.

III. The Savior Solution: A Value-Driven, Sustainable Approach

Savior’s integrated strategy addresses the dual public health and environmental crises by synthesizing advanced technology with radical affordability and a commitment to sustainable operations. The brand offers a three-pronged solution:

1. **Purity Assurance:** Employing advanced multi-stage filtration (RO/UV) to reliably eliminate the systemic health threats prevalent in groundwater, such as arsenic and heavy metals.
2. **Affordability:** Utilizing a lean production setup, maximized operational efficiency (optimized RO recovery, solar power integration), and a penetration pricing model to

achieve a price point 20-30% lower than established competitors.
3. **Eco-Responsibility:** Committing to durable and plant-based packaging, coupled with community engagement and internal sustainable operations, to actively minimize the brand's environmental footprint.

A. Unique Value Proposition (UVP)

Savior's unique value proposition is the seamless merger of high-quality purification technology with unparalleled consumer affordability and environmental accountability. This positioning empowers the price-sensitive mass market to make a responsible consumption choice without incurring a financial penalty.

The primary competitive edge is offering the **Best Value for Money**. By selling the core 500ml product at 14–16 BDT, Savior directly challenges the market assumption that certified purity must come at a premium price, significantly undercutting Kinley (20–22 BDT) and Mum (18–20 BDT).
This affordability is paired with a guarantee of **Unbeatable Purity**. Certification and transparent testing protocols are leveraged to build the consumer trust currently monopolized by premium brands.

Finally, the brand delivers **Eco-Friendly Empowerment**. The strategy enables the mass market, which faces acute financial pressures, to participate in environmental stewardship through conscious packaging choices and clear instructions for disposal and reuse. This dual promise allows Savior to appeal to both the financially constrained consumer seeking essential safe water and the growing segment of environmentally conscious buyers.

B. Mitigation of Perceived Lower Quality

A significant challenge faced by budget brands, such as Fresh (15–18 BDT), is the consumer perception of lower quality.[1] Given that Savior's price point (14–16 BDT) is positioned at the budget end of the market, the brand must deploy a strategic mechanism to negate the implicit association between low price and low quality.

The strategy hinges on aggressively emphasizing two differentiating factors: **advanced technology and transparency**. Savior must market its purification process—detailing the multi-stage RO, carbon, and UV sterilization steps—to clearly demonstrate technical superiority over undifferentiated budget offerings. By highlighting certification, process rigor, and advanced system engineering (e.g., CCPP optimization), Savior constructs a narrative of meticulous quality control that fundamentally contradicts the notion of low quality. This emphasis on technical excellence and transparency ensures that the consumer perceives the low price as a result of operational efficiency and ethical volume pricing, rather than a compromise on product safety or purity, thereby successfully overcoming the critical

weakness faced by competitors like Fresh.

IV. Product Excellence and Purification Technology

A. Product Description and Phased Rollout

The Savior product line is systematically designed to meet varying consumer needs, spanning impulse purchases, daily hydration, and B2B/household bulk requirements.

The **Initial Launch Sizes (Phase 1)** include 250ml bottles, optimized for impulse buys, students, and street vendors; 500ml bottles, serving as the core mass-market daily carry product; and 1 Liter bottles, suitable for convenience and small office use.

The **Expansion Sizes (Phase 2)**, planned as part of the 5-Year Growth Roadmap, will introduce 2 Liter and 5 Liter bottles. These large formats are essential for penetrating the lucrative household and office markets, providing consumers with greater bulk savings and further integrating Savior into B2B supply chains.

Beyond safety, the product focuses on superior taste. The multi-stage filtration process effectively removes Total Dissolved Solids (TDS) and harmful contaminants. Post-RO, the water will undergo a re-mineralization process to introduce essential minerals, optimizing the final taste profile and ensuring the water is palatable and refreshing, distinguishing it from water that can taste flat due to overly aggressive purification.

The table below illustrates the pricing and strategic role of each product size in the phased rollout:

Table 1: Savior Product Line Rollout and Targeted Pricing Structure

Bottle Size	Target Market Segment	Cost/Bottle (BDT)	Selling Price (BDT)	Competitive Advantage
250 ml	Impulse, Students, Street Vendors	7.00	8.50	Immediate Affordability, High Volume Potential
500 ml	Daily Consumption, General Mass Market	9.80	12.30 – 14.00	Core Value Proposition, Competitive Pricing Leader

1 Liter	Convenience, Small Office Use	13.00	16.00 – 19.00	High Utility, Strong Unit Cost Efficiency
2 Liter (Phase 2)	Residential, Small Office B2B	21.00	26.00 – 27.00	Household Convenience, Bulk Savings
5 Liter (Phase 2)	Large Household, Corporate B2B	40.00	50.00 – 60.00	Office Water Cooler Replacement, Maximum Value

B. Purification Technology: Multi-Stage Excellence and Efficiency

Savior utilizes a rigorous, multi-stage purification protocol that goes beyond basic filtration to ensure unparalleled safety and purity, directly neutralizing the endemic threats found in local source water.

B.1. Detailed Purification Process

The purification line begins with the **Pre-Treatment and Filtration Stage**. This involves using depth filters to remove large particles, sediments, and rust greater than 5 microns. This initial step is operationally vital, acting as a barrier that protects the more sensitive downstream RO membranes from premature wear and blockages, potentially cutting maintenance costs by approximately 35%.

This is followed by **Activated Carbon Filtration**, a step essential for eliminating chlorine and organic compounds. Chlorine negatively affects membrane longevity and, if left untreated, imparts an unpleasant taste and odor to the final product.

The core of the system is the **Multi-Stage Reverse Osmosis (RO)** process. This involves using high-pressure pumps to force water through a semi-permeable membrane, achieving exceptional contaminant rejection. RO systems are capable of removing up to 99% of Total Dissolved Solids (TDS), including dissolved salts, heavy metals (such as arsenic and manganese), viruses, and bacteria. This process is the necessary technical solution for ensuring the water is safe from the widespread arsenic contamination crisis in Bangladesh.

Finally, **Post-Treatment and Sterilization** completes the process. Water is polished using absolute rated 1 micron filters to ensure clarity. A dedicated UV sterilization or ozonation unit is employed immediately prior to bottling. This stage provides the final kill step, guaranteeing

bacteria and pathogen-free water, achieving 99.99% removal efficiency before the product reaches the consumer.

B.2. Optimized System Engineering to Reduce Water Wastage

A critical operational challenge for any RO plant is balancing efficiency with sustainability, specifically managing the **Recovery Rate**—the percentage of feed water converted into purified permeate versus concentrate (wastewater). Standard industrial RO systems can generate significant brine, increasing the water footprint and operational costs. Savior's commitment to sustainability necessitates an aggressive focus on maximizing the recovery rate.

The ultimate limit to recovery is determined by the management of scaling and fouling, which occurs when high concentrations of dissolved salts (such as calcium carbonate and silica) precipitate in the reject stream as recovery rates increase. To safely push the recovery rate higher, Savior will utilize advanced water quality analysis that moves beyond traditional Langelier Saturation Index (LSI) or Silt Density Index (SDI). Instead, the system employs **Calcium Carbonate Precipitation Potential (CCPP)** analysis.

CCPP provides a quantitative measurement of the actual precipitation tendency, which is superior to the qualitative indices (LSI/SDI) typically used to predict scaling potential. By precisely quantifying the scaling risk, Savior can optimize chemical pre-treatment and membrane flushing protocols, thereby stabilizing system performance and membrane lifespan while achieving a significantly higher permeate recovery rate. This optimized system engineering targets an aggressive recovery rate of 75-80%, dramatically reducing water consumption and ensuring that the brand's environmental claims are supported by measurable operational efficiency gains.

V. Sustainable Packaging Strategy

Savior's commitment to sustainability is manifested in a nuanced packaging strategy that pragmatically addresses the infrastructural limitations of Bangladesh's waste management system, focusing on promoting reuse and durability.

A. Features of the Eco-Friendly Bottle

A.1. The Blended Material Approach

Savior utilizes a carefully selected blend of materials tailored to the specific use case and necessary durability.
For smaller, high-volume, single-use formats (250ml and 500ml), the brand will incorporate **Polylactic Acid (PLA)**, a plant-based plastic. PLA is a 100% compostable and recyclable

material derived from non-GMO plants, eliminating the reliance on petroleum resources.

For larger format bottles (1L, 2L, and 5L), which are intended for multi-use household and B2B applications, the focus shifts to durability and superior recyclability where collection systems exist. These bottles will utilize high-density PET (Polyethylene Terephthalate) or HDPE (High-Density Polyethylene). PET is favored for its transparency and robustness in water bottling, and both PET and HDPE are 100% recyclable, playing a significant role in the circular economy when properly managed. HDPE is often easier to recycle into non-food products.

A.2. Addressing the Compostability Paradox

While the use of PLA is a positive move toward reducing reliance on fossil feedstocks, a crucial limitation must be recognized and addressed transparently: the lack of widespread industrial composting infrastructure in Bangladesh. Without access to the high-heat, controlled conditions of industrial composting facilities, PLA in a typical Bangladeshi landfill or waterway will behave much like conventional plastic.

Therefore, Savior's environmental communication must strategically shift its primary emphasis from theoretical 'compostability' to the demonstrable benefit of **'Durability and Active Reuse,'** particularly for the larger PET and HDPE bottles. This strategic adjustment is essential to maintain credibility. The brand's immediate impact must be measured by how effectively it reduces the net consumption of single-use virgin plastic by encouraging consumers to retain and refill the durable containers multiple times.

B. Instructions on Proper Disposal and Long-Term Reuse

Given the low consumer awareness regarding the health and environmental risks of plastic (only 18.4% of urban consumers are aware) and the resulting careless disposal, consumer education is a core element of Savior's sustainability commitment.

Packaging will feature clear, multi-lingual instructions and potentially QR codes linking to educational content. These instructions will emphasize:

1. **Reuse:** Strongly encouraging the consumer to refill and reuse the durable PET/HDPE bottles for long-term use, maximizing their lifecycle value.
2. **Segregation:** Providing guidance on proper disposal, advising consumers to separate bottles from general municipal waste. Segregating by material type (PET vs. other plastics) simplifies the process for the rudimentary local recycling sector and prevents contamination.
3. **Future Collection Program:** Committing to the development and piloting of a brand-specific bottle collection and buy-back program (incentivizing return of durable bottles) once production volume and distribution reach a scale that justifies the logistical investment.

VI. Market Opportunity and Competitive Dynamics

A. Target Market and Consumer Profile

Savior targets the vast segment of the Bangladeshi market defined by acute price sensitivity combined with a non-negotiable need for certified water safety.

The **Primary Segment (B2C)** comprises price-sensitive urban consumers, including students, workers, and low-to-lower-middle-income households, particularly within metropolitan areas like Dhaka, Narayanganj, and Gazipur. These consumers are heavily impacted by the rising costs of essential commodities (e.g., rice, oil, lentils have increased by 10–25% year-on-year). Providing safe water at a 20-30% discount is critical for this demographic.

The **Secondary Segment (B2B/Bulk)** includes small offices, universities, and large urban households that rely on bulk water supply, addressed by the planned introduction of 2L and 5L formats.

Surveys indicate that consumers prioritize hygienic bottled water and, crucially, are **open to switching brands in pursuit of better quality or price**. This price elasticity and willingness to switch validate Savior's strategy of combining certified purity with market-leading affordability.

B. Market Demand & Opportunity

The bottled water industry in Bangladesh is structurally poised for continued expansion. Despite short-term contractions caused by events like the COVID-19 pandemic, the market is projected to reach approximately US$1.8 billion in revenue within the coming years, with an expected Compound Annual Growth Rate (CAGR) of 3.14% between 2024 and 2027. This growth is structurally supported by accelerating urbanization, rising disposable incomes, and increasing public health awareness stemming from documented water contamination crises.

The market exhibits a significant **gap** that Savior is designed to fill. Existing premium brands (Kinley) offer trust but are prohibitively expensive for the mass market.[1] Existing budget brands (Fresh) offer low prices but suffer from perceptions of compromised quality.[1] The market lacks a product that successfully synthesizes **certified, advanced purification** (necessary to address arsenic/heavy metal risks) with **extreme affordability** and a proactive **eco-responsibility** stance. Savior's value/sustainability positioning provides a unique market entry point to capture the volume-driven core of this expanding sector.

C. Competitor Analysis and Competitive Advantage

Savior's strategy is explicitly designed to capitalize on the financial and sustainability

weaknesses of established incumbents. The competitive landscape for 500ml bottles demonstrates clear positioning:

Table 2: Competitive Landscape Comparison (500ml Price Benchmark)

Brand	Market Position	Avg 500ml Price (BDT)	Core Strength	Strategic Weakness
Kinley	Premium	20 – 22	Strong trust, Perceived Quality	High Cost Barrier to Entry for Mass Market
Mum	Mass Market	18 – 20	Wide, Saturated Distribution	Lack of Purity Differentiation, Brand Saturation
Fresh	Budget	15 – 18	Low Initial Price Point	Perceived Lower Quality, Undifferentiated Plastic Use
SAVIOR	**Value/Sustainable Leader**	**14 – 16**	**Unbeatable Value, Certified Purity, Eco-Branding**	**Requires awareness and initial distribution build-up** [1]

C.1. Strategic Exploitation

Savior gains a significant competitive edge through price disruption. At 14–16 BDT, Savior is priced 20% to 36% lower than the premium competitor Kinley, and 11% to 22% lower than the mass-market leader Mum. This aggressive pricing immediately addresses the high price sensitivity of the target consumer base.

Furthermore, Savior attacks the core weaknesses of budget competition. By positioning its product as having "Unbeatable Purity" supported by advanced RO/UV technology and transparent quality testing, Savior directly counters Fresh's weakness of "Perceived lower quality".

Finally, no major competitor has aggressively adopted a measurable sustainable packaging and operations model (e.g., solar power, optimized RO). Savior's eco-friendly branding and operational commitment provides a crucial ethical differentiator that resonates with growing, though currently small, segments of environmentally aware consumers, while simultaneously setting a high bar for future competitors.

VII. Commercialization and Growth Strategy

A. Pricing Strategy: Penetration for Volume

Savior employs a high-volume penetration pricing model as the cornerstone of its commercial strategy. The objective is to rapidly secure substantial market share by offering certified, high-quality water at 20-30% below the market average. This strategy is designed to immediately incentivize brand switching and establish consumer loyalty in a price-elastic market.

The sustainability of this penetration pricing model relies entirely on achieving scale to generate operational efficiencies. The brand's lean production setup, incorporating solar power and optimized high-recovery RO systems, contributes to a lower unit cost. However, the initial financial analysis illustrates that fixed operating costs, totaling 28,500 BDT per month (including rent, utilities, labor, and transportation), pose the immediate obstacle to profitability.

The test phase production volume of 7,000 units generates only 18,000 BDT in variable profit, resulting in a net monthly loss of 10,500 BDT. This structural observation mandates that the operational capacity must be quickly increased to amortize the fixed costs across a larger volume base. The plan for Year 1 is to increase production volumes and is predicated on the expectation that break-even will be achieved after doubling the current scale, requiring a volume of approximately 14,000 units per month. The pricing strategy, therefore, is not merely a marketing tool but a direct engine for achieving the necessary volume threshold for financial viability.

B. Distribution Plan

The distribution strategy is phased to match operational scale and investment capacity.
In **Phase 1**, the focus is on achieving immediate, high-visibility placement. This involves securing local partnerships and targeting locations frequented by the primary price-sensitive market: shops, university campuses, tea stalls, and offices. Leveraging existing small-scale distributor networks ensures rapid, localized market entry.

In **Phase 2**, as production expands to include 2L and 5L formats, the strategy will incorporate

a dedicated **Online Ordering and Delivery** network. This move is crucial for penetrating the B2B (office) and residential household markets. Delivery logistics will prioritize **Eco-Friendly Methods**, deploying electric vans and bicycles for last-mile delivery in densely populated urban zones like Dhaka. This commitment reinforces the brand's environmental claims and provides a measurable reduction in logistical carbon emissions.

C. Marketing Strategy

The marketing strategy is designed to build rapid brand awareness and establish a strong association between the Savior name, certified purity, and unparalleled value, overcoming the initial challenge of "Awareness needed".

The brand will utilize a concise and impactful **Brand Slogan** that captures its dual purpose **Social Media Campaigns** will be central to building trust. Campaigns will prioritize visual transparency, showcasing the advanced, multi-stage RO and UV purification process. This visual evidence of technical rigor is designed to counter any potential public association between the low price point and low product quality. Digital platforms will also be used extensively for consumer education, detailing the health risks of contaminated water and providing clear instructions on the benefits of reusing the durable packaging.

Influencer Promotions and Partnerships will focus on collaborating with respected local environmental and public health advocates rather than conventional celebrity endorsements. This aligns the brand authentically with its ethical mission.

Loyalty Programs will be implemented to drive long-term reuse behavior. This could include targeted refill discounts or structured return schemes, particularly for the durable PET/HDPE bottles, incentivizing consumers to reduce their disposal frequency and build brand allegiance.

Finally, Savior will demonstrate genuine **Social Commitment** by participating visibly in environmental events, such as sponsoring community clean-up drives and running water safety awareness campaigns.

VIII. Production and Operations Framework

Operational excellence, coupled with sustainable practices, is the core mechanism allowing Savior to deliver premium quality at a highly competitive price point.

A. Production / Operations Overview

The production facility is engineered for high throughput and rigorous quality control. It utilizes integrated, advanced multi-stage purification units encompassing UV, RO, and post-filtration mineral systems.

Quality Control is paramount, ensuring consistent delivery of the "Unbeatable Purity" promise. This is managed through the integration of smart sensors deployed across the production line to monitor critical metrics like TDS levels, differential pressure, and turbidity in real-time. Rigorous daily quality testing protocols are established and implemented before any finished product is cleared for delivery. The unique application of CCPP analysis (discussed in Section IV.B.2) is a core component of maintaining quality by optimizing the RO system performance.

A foundational element of the operational strategy is the strategic commitment to investing profits into and installing **Solar-Powered Production Facilities**. This investment serves multiple critical business objectives: it hedges against the instability and rising costs of the national electricity grid, stabilizing long-term operational expenses, and provides a powerful, measurable metric for the brand's reduced carbon footprint, reinforcing its environmental leadership.

The production line emphasizes high automation to support a lean operational model. This efficient bottling line minimizes the dependency on extensive manual labor, aligning with the low fixed cost structure necessary to support the penetration pricing strategy.

B. Organizational Structure

The initial organizational structure is designed to be lean and focused, prioritizing the key functions necessary for rapid scaling, particularly quality assurance and market penetration.

- **CEO/Managing Director:** Responsible for setting the strategic direction, securing investment, and managing critical external stakeholder relations.
- **Operations Manager:** Oversees day-to-day production, logistics, supply chain management, and efficient maintenance of the sophisticated RO/UV systems.
- **Quality Officer (QC):** This is a mission-critical role. The QC officer manages daily purity testing, ensures compliance with national and international drinking water standards, and administers the advanced sensor and CCPP monitoring systems to guarantee the brand's core promise of safety.
- **Marketing & Distribution Manager:** Tasked with executing the penetration marketing campaigns, managing social media presence, coordinating local distribution partnerships, and overseeing the rollout of the eco-friendly delivery fleet.
- **Finance Officer:** Manages cost control, tracks profitability against volume targets, and executes the strategic reinvestment plan (solar power, technology upgrades).
- **Delivery/Production Team:** The essential local workforce supporting bottling,

packaging, and logistics fulfillment (initial labor costs are intentionally lean).[1]

IX. Financial Projections and Investment Viability

A. Financial Projections: Path to Profitability

Savior's financial model demonstrates that viability is inherently linked to achieving aggressive volume scaling to absorb fixed operating costs quickly.

A.1. Initial Cost Structure and Test Phase Results

The test phase serves as a baseline for the challenge posed by fixed costs.

The initial production (7,000 total units per month, comprising 3,000 units of 250ml, 3,000 units of 500ml, and 1,000 units of 1L) generates a total revenue of 88,000 BDT and a total variable profit (after material and variable costs) of 18,000 BDT.

The fixed operating costs, budgeted at 28,500 BDT per month, include Space Rent (10,000 BDT), Electricity/Water (3,500 BDT), Labor for one helper (8,000 BDT), Transportation (5,000 BDT), and Misc/Packaging Loss (2,000 BDT).

The resulting Net Monthly Profit for the Test Phase is calculated as 18,000 BDT (Variable Profit) minus 28,500 BDT (Fixed Costs), leading to an expected test-phase loss of **-10,500 BDT**.

A.2. Year 1 Outlook and Break-Even Analysis

The strategic objective for Year 1 is to increase production volumes dramatically to reduce the cost per unit and achieve financial break-even.

The break-even point is achieved when the total variable profit equals the fixed costs of 28,500 BDT. Based on the variable profit margin structure, this necessitates doubling the current 7,000 unit production scale to approximately 14,000 units per month. The penetration pricing model is intentionally aggressive to drive the consumer volume required to cross this break-even threshold within the first 6 to 9 months of operation.

Table 3: Financial Scaling Requirement: Path to Break-Even (Year 1 Target)

Metric	**Test Phase (7,000 Units)**	**Fixed Operating Costs**	**Break-Even Target (Estimate)**	**Targeted Year 1 Profitability**
Monthly Revenue (BDT)	88,000	N/A	$\approx$150,000	$\ge$250,000
Total Variable Profit (BDT)	18,000	N/A	28,500	$\ge$70,000
Total Fixed Costs (BDT)	28,500	28,500	28,500	$\approx$30,000
Net Monthly Profit (BDT)	-10,500 (Loss)	N/A	**0**	$\ge$40,000
Required Production Scale (Units)	7,000	N/A	$\approx$14,000	$\ge$20,000

A.3. Five-Year Growth Roadmap and Reinvestment

The 5-Year Growth Roadmap centers on expanding production capacity, distribution zones, and introducing the higher-margin 2L and 5L bottles for the household and office markets. These larger formats significantly increase the profit per transaction and stabilize the brand's revenue base through B2B partnerships.

Profits generated during the scaling phase will be strategically reinvested to reinforce the brand's long-term competitive advantage: (1) Investment in **Solar Power Infrastructure** to stabilize utility costs and enhance ESG credentials; (2) Upgrading **Advanced Bottling Technology** to further reduce unit costs and support potential changes in eco-packaging materials; and (3) Expanding the dedicated, eco-friendly delivery network.

The projected profitability following the successful integration of Phase 2 products illustrates the immense scaling potential:

Table 4: Projected Annual Profitability (Scaling Model - Phase 2 Targets)

Bottle Size	Monthly Unit Target (Scaled)	Profit/Bottle (BDT)	Monthly Profit (BDT) Target	Annual Profit (BDT) Target
250 ml	10,000	1.50	15,000.0	180,000.0
500 ml	10,000	2.50	25,000.0	300,000.0
1 L	10,000	3.00	30,000.0	360,000.0
2 L (Phase 2)	10,000	5.00	50,000.0	600,000.0
5 L (Phase 2)	10,000	10.00	100,000.0	1,200,000.0
Total Target (Phase 2)	50,000 units	N/A	**220,000.0**	**2,640,000.0**

B. Viability Analysis

The fundamental viability of the Savior model rests on its operational efficiency offsetting the aggressive pricing structure. The ability to maintain prices 20–30% lower than competitors is sustainable because the cost structure is designed to be proportionally leaner. This efficiency is achieved through the three pillars of operational advantage: high volume sales resulting in reduced fixed cost per unit, predictable utility cost management via solar power integration, and sophisticated raw material savings through optimized RO permeate recovery (using CCPP analysis). These efficiencies transform the initial low variable profit per bottle into scalable net profit once sufficient volume is attained to cover overheads.

X. Integrated Social and Environmental Impact

Savior is structured as a profitable entity whose commercial success inherently generates positive externalities, making it a compelling investment for stakeholders focused on Environmental, Social, and Governance (ESG) compliance.

A. Environmental Impact (E)

The brand directly addresses the core challenges of environmental degradation in Bangladesh.

Plastic Waste Reduction: By adopting a strategy that emphasizes durable PET/HDPE for larger volumes and promoting consumer reuse through educational programs and loyalty incentives, Savior actively works to minimize the brand's contribution to the 78.6% of single-use bottles that end up polluting the environment annually.

Carbon Footprint Reduction: The commitment to solar-powered production facilities significantly reduces reliance on fossil fuel energy sources for manufacturing, leading to a measurable decrease in the brand's overall operational carbon emissions. Furthermore, the commitment to eco-friendly logistics, utilizing electric vans and bicycles for distribution, minimizes transportation-related greenhouse gas emissions.

Water Conservation: Through the implementation of advanced RO system engineering, particularly the use of CCPP analysis to maximize permeate recovery, Savior targets a high efficiency rate (75–80%). This strategic focus ensures responsible consumption of sourced water, minimizing the industrial wastewater footprint.

Community Programs: Savior institutionalizes its environmental commitment through active, local interventions, including sponsoring community clean-up drives targeting highly polluted waterways like the Buriganga River and running local tree planting and environmental awareness initiatives.

B. Social Impact (S)

Savior's social impact is centered on improving public health outcomes and fostering local economic development.

Access to Clean Water: By providing reliably certified, arsenic and heavy-metal-free drinking water at a price accessible to the low-to-lower-middle-income majority, Savior directly addresses a critical public health failure in Bangladesh. This provision of safe water contributes substantially to the welfare of price-sensitive households that currently lack sustainable, safe alternatives.

Job Creation and Skill Development: The expansion of production and the establishment of a dedicated delivery network create new local employment opportunities in advanced manufacturing, technical quality control, and logistics. The workforce will benefit from training in modern purification and quality management systems, enhancing local skill sets.

Health and Safety: By offering a sustainable, safe alternative to contaminated sources, Savior contributes directly to the national mitigation strategy against chronic diseases associated with poor water quality, including arsenicosis and various cancers.

XI. Conclusion and Call to Action

Savior Bottled Water presents a strategically sound, high-growth investment opportunity that capitalizes on severe market failures in public health and environmental stewardship in Bangladesh. The brand successfully bridges the gap between premium quality and mass-market affordability through technological innovation (multi-stage RO/UV, CCPP-optimized recovery) and operational discipline (solar power integration, lean cost structure).

The financial analysis confirms that the aggressive penetration pricing model is a viable long-term strategy, contingent upon achieving the requisite volume scaling (doubling initial capacity) in Year 1 to absorb fixed overheads. The phased roadmap, which introduces high-margin 2L and 5L formats, projects significant future profitability, targeting annual profits exceeding 2.6 million BDT upon full scale operation.

Savior offers investors and partners a dual return: robust financial growth potential in a rapidly expanding market, coupled with demonstrable, measurable social and environmental returns aligned with global ESG frameworks.

We urge potential investors and strategic partners to support the initial launch and the accelerated scaling of Savior. Investment capital is earmarked for the immediate procurement and installation of high-efficiency production technology, solar power infrastructure, and the foundational development of the targeted distribution network essential for achieving critical Year 1 volume targets. Join the initiative to establish Savior as the market leader in affordable, safe, and sustainable bottled water in Bangladesh, delivering **Pure Water, Pure Value, and a Pure Choice** for the community and the environment.

References:

1. How to Maximize the Recovery Rate of Your RO System? - Rotek, accessed November 9, 2025, https://www.rotekwatershop.com/post/howtomaximizerosystemrecoveryrate
2. Water supply and sanitation in Bangladesh - Wikipedia, accessed November 9, 2025, https://en.wikipedia.org/wiki/Water_supply_and_sanitation_in_Bangladesh
3. SURFACE AND GROUND WATER QUALITY REPORT 2023, accessed November 9, 2025, https://doe.portal.gov.bd/sites/default/files/files/doe.portal.gov.bd/publications/edc0f4f2_8346_4f58_bdd1_72978984ba3e/2024-08-27-08-00-a698975451058b1fc83366d454942747.pdf
4. Arsenic contamination in groundwater in Bangladesh: implications and challenges for healthcare policy - PubMed Central, accessed November 9, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC6281155/
5. Is the groundwater of Dhaka city, Bangladesh contaminated with naturally occurring potential toxic elements? - Frontiers, accessed November 9, 2025, https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2024.1514154/full
6. Bangladesh disposes around 3.5b single-use plastic bottles annually: Study, accessed November 9, 2025, https://www.tbsnews.net/bangladesh/environment/bangladesh-disposes-around-35b-single-use-

plastic-bottles-annually-study
7. BD uses 3.84b plastic bottles a yr: Study | The Financial Express, accessed November 9, 2025, https://thefinancialexpress.com.bd/economy/bd-uses-384b-plastic-bottles-a-yr-study
8. Meeting Bangladesh's Plastic Challenge through a Multisectoral Approach - World Bank, accessed November 9, 2025, https://www.worldbank.org/en/news/feature/2021/12/23/meeting-bangladesh-s-plastic-challenge-through-a-multisectoral-approach
9. Microplastics pollution in Bangladesh: a decade of challenges, impacts, and pathways to sustainability - Oxford Academic, accessed November 9, 2025, https://academic.oup.com/ieam/advance-article/doi/10.1093/inteam/vjaf108/8231076
10. Bangladesh's Plastic Tide: A Nation Grappling with a Mounting Crisis, accessed November 9, 2025, https://ishr.org/bangladeshs-plastic-tide-a-nation-grappling-with-a-mounting-crisis/
11. Dhaka Sitting on a Plastic Bomb: Issues and Concerns around Waste Governance, Water Quality, and Public Health - MDPI, accessed November 9, 2025, https://www.mdpi.com/2673-4834/3/1/2
12. Polyethylene Terephthalate (PET) Recycling in Bangladesh: Current Status and Future Sustainability Prospects - IEOM Society International, accessed November 9, 2025, https://ieomsociety.org/proceedings/bangladesh2024/148.pdf
13. Polyethylene Terephthalate (PET) Recycling in Bangladesh: Current Status and Future Sustainability Prospects - ResearchGate, accessed November 9, 2025, https://www.researchgate.net/publication/387333988_Polyethylene_Terephthalate_PET_Recycling_in_Bangladesh_Current_Status_and_Future_Sustainability_Prospects
14. RO Water Treatment Plant: The Ultimate Purification Solution - Romegamart, accessed November 9, 2025, https://romegamart.com/blog-details/ro-water-treatment-plant-the-ultimate-purification-solution/
15. PLA bottles, a plant-based, compostable, recycable bioplastic bottle., accessed November 9, 2025, https://plabottles.eu/
16. HDPE vs PET Bottles: Which Is Right for Your Product? - PT Bioplast Unggul, accessed November 9, 2025, https://bioplast.co.id/en/news/differences-between-hdpe-and-pet-bottles/
17. Water Production Line: From Raw Water Filtration to Bottling in One Flow, accessed November 9, 2025, https://www.xmfiller.com/blog/water-production-line-from-raw-water-filtration-to-bottling-in-one-flow
18. Performance Metrics for Optimizing Reverse Osmosis Systems - AXEON Water, accessed November 9, 2025, https://www.axeonwater.com/blog/key-membrane-performance-metrics-for-optimizing-reverse-osmosis-systems/
19. Packaging - Bottled Water | IBWA, accessed November 9, 2025, https://bottledwater.org/packaging/
20. 3 Things You Didn't Know About Sustainable Packaging (Bioplastics, PLA and Paper), accessed November 9, 2025, https://varden.com.au/3-things-about-sustainable-packaging-and-pla/
21. The bottled water industry in Bangladesh - Emerging Credit Rating Ltd, accessed November 9, 2025, https://emergingrating.com/monthly-magazine-april-2024-5-3-2/
22. Publication: Bangladesh - Prices of Essential Commodities : Recent Trends, Underlying Factors and Policy Options, accessed November 9, 2025, https://openknowledge.worldbank.org/entities/publication/b83bb8ee-15f1-5045-8a0f-b3f026b4a74a
23. Consumers struggle with soaring commodity prices ahead of Ramadan - Dhaka Tribune, accessed November 9, 2025, https://www.dhakatribune.com/bangladesh/374250/consumers-struggle-in-bangladesh-with-soaring
24. Pure Drinking Water Industry in Bangladesh | PRITI Research and Consultancy Limited, accessed November 9, 2025, https://pritiresearch.com.bd/pure-drinking-water-industry-in-bangladesh/